Experiment - 9

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Subject Name: System Design **Subject Code:** 23CSH-314

Aim:

To design and analyze a scalable online movie ticket booking system that allows users to browse movies, select seats, and make payments in real time, ensuring high availability, concurrency handling, and low latency.

Objectives:

1. To understand system design principles for large-scale booking systems.
2. To design APIs for movie listing, seat booking, and payment services.
3. To identify functional and non-functional requirements.
4. To handle concurrency issues during seat booking.
5. To design a high-availability and scalable system architecture.

Procedure-

1. Identify functional requirements of the movie ticket booking system.
2. Define non-functional requirements such as scalability, availability, and latency.
3. Analyze concurrency challenges in seat booking.
4. Identify core entities such as User, Movie, Theatre, Show, Seat, and Booking.
5. Design the system architecture using Draw.io.
6. Validate the design for booking, payment, and seat allocation scenarios.

Functional Requirements –

- User should be able to search events by title, location, and date.
- User should be able to view event details including description, metadata, and seat availability.
- User should be able to select seats and book tickets.

Non-functional Requirements-

- System should support up to 100 Million Daily Active Users (DAU).
- System should ensure high availability for search and event viewing.
- System should ensure strong consistency during ticket booking (no double booking).
- System should provide low latency responses for better user experience.

Core Entities:

- **User** → userId, name, email
- **Event** → eventId, title, date, location
- **Venue** → venueId, name, location
- **Seat** → seatId, eventId, status
- **Booking** → bookingId, userId, eventId, status
- **Payment** → paymentId, bookingId, amount, status

Api Design:

- GET /events/search → search events by title, location, date
- GET /events/{eventId} → get event details + seats
- POST /bookings/reserve → reserve selected seats
- POST /bookings/confirm → confirm booking
- POST /payments → process payment
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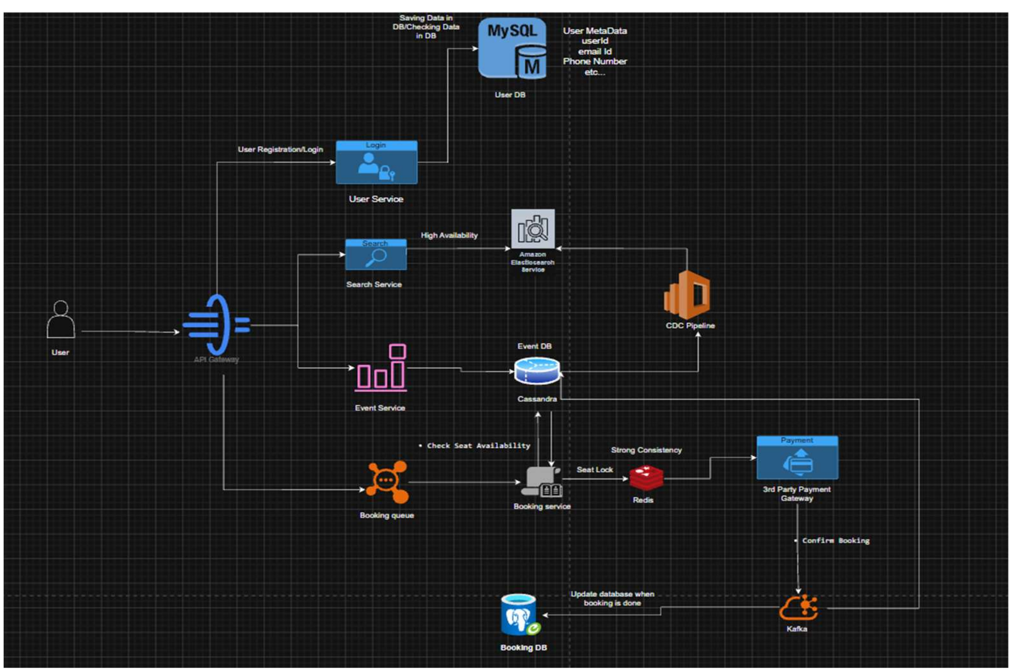
HLD:



Outcome / Result -

The online movie ticket booking system was successfully designed with features like movie browsing, seat selection, secure payments, and concurrency handling, ensuring scalability, high availability, and efficient real-time booking under heavy load conditions.